

ExCP: Extreme LLM Checkpoint Compression via Weight-Momentum Joint Shrinking

Wenshuo Li; Xinghao Chen; Han Shu; Yehui Tang; Yunhe Wang

01 We hit a bug mid-run: checkpoint bloat blocks fast recovery

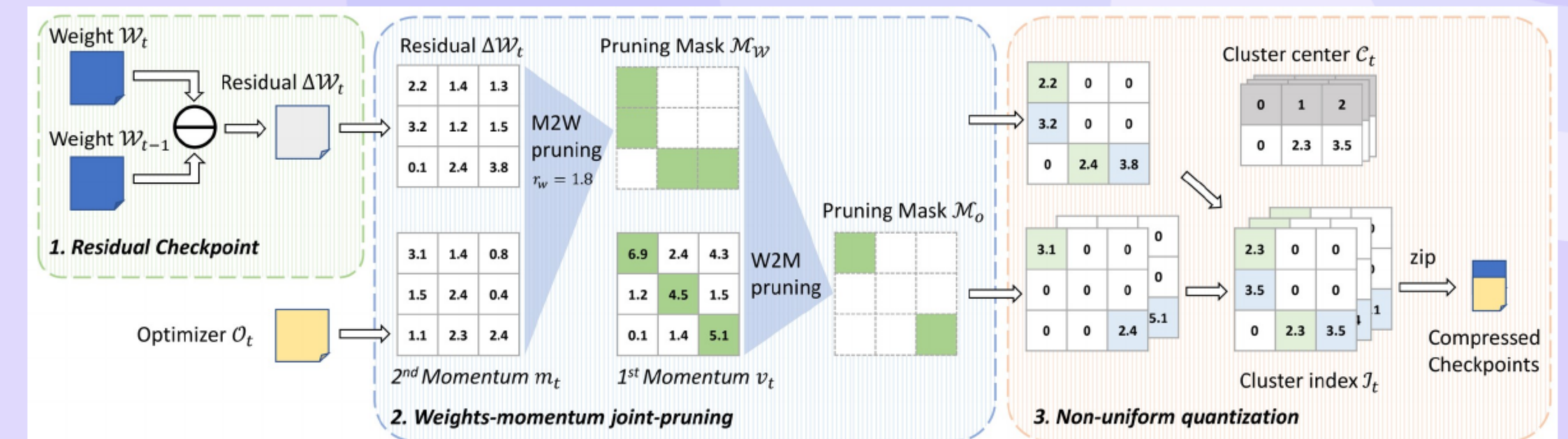
- We train 12B models; at step 120k we must roll back safely across nodes.
- Each save is huge (Pythia-12B >24 GB); 154 saves ≈11 TB; moments roughly double storage.
- We ask: Can we store many more checkpoints without hurting resume fidelity?

02 Problem & Gaps in Prior Work

Method	M2W pruning	W2M pruning	Top-1 Accuracy(%)	CR(Weights)	CR(Weights & Momentum)
baseline			71.36	1	1
CNR+			71.57	7.82	1.41*
QD+			71.24	16.31	1.45*
DynaQuant			71.82	26.19	1.47*
ExCP(Ours)			71.51	-	19.88
ExCP(Ours)	✓		71.53	-	25.54
ExCP(Ours)		✓	71.80	-	22.76
ExCP(Ours)	✓	✓	71.69	-	35.21

- We need frequent checkpoints, but weights+moments overwhelm storage/I/O; generic/inference methods don't fit training checkpoints.
- Prior work often quantizes only weights, ignores temporal redundancy, and omits optimizer states—hurting resumes.
- We need a checkpoint-specific method; we propose ExCP; we show higher compression without sacrificing resume fidelity.
- Table: prior methods vs. ours—only ExCP jointly handles weights and moments with higher CR.

03 ExCP at a Glance: Residuals + Joint Shrinking + Non-uniform Quantization (Conceptual)



- Residual encoding: store weight deltas; expose sparsity.
- Joint shrinking: prune weights+moments together.
- Non-uniform quantization: data-driven bins.
- Residuals help: small, structured deltas compress.
- Joint treatment: align weights and moments.
- Adam intuition: drop tiny states; preserve convergence.
- Results: up to 70× with near-lossless resumes.

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Experimental Setup: Models, Tasks, Protocol (At a Glance)

	zip	rar	rar4	bz2	7z
size	73M	70M	69M	64M	61M

- Models: ViT-L/32; Pythia-410M; PanGu π -1B/7B.
- Benchmarks: ImageNet-1K; HellaSwag, ARC-e, PIQA, C3, CSL, LAMBADA; PanGu π suite.
- We measure fidelity by resuming from compressed vs. original checkpoints and comparing outcomes.
- Table: final archive sizes—7z yields the smallest file.

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Main Results Across Models (Pythia, ViT, PanGu π)

Model	Method	Size	Tasks						
			hellaswag	arc-e	piqa	C3	cs1	lambada	Avg
Pythia-410M	Original model	4.53G	32.52	35.80	62.13	37.21	53.75	37.22	43.11
	Residual+7Zip	3.40G	32.52	35.80	62.13	37.21	53.75	37.22	43.11
	ExCP (Ours)	0.06G	31.95	37.04	62.62	36.22	52.50	37.24	42.93

- On Pythia-410M, we achieve ~70 $\times$ (4.53 GB $\rightarrow$ 0.06 GB) with near-lossless average across tasks.
- On ViT-L/32, we retain ImageNet-1K accuracy at >30 $\times$ compression; counting optimizer states widens our margin over weight-only baselines.
- On PanGu π -1B/7B, we reach ~25 $\times$ with negligible change after resume.
- Table: Pythia-410M—70 $\times$ reduction with near-lossless average.

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Ablation Insights: where gains come from (residuals, joint pruning, non-uniform quantization)

method			Size	Avg Acc
residual	prune	quant		
✓			4070M	43.11
			3484M	43.11
	✓		324M	29.95
		✓	492M	40.17
✓	✓		276M	42.92
✓		✓	493M	42.94
✓	✓	✓	61M	42.93

- Residuals expose sparse, structured updates $\rightarrow$ stabilize accuracy.
- Joint shrinking plus non-uniform quantization compounds gains while preserving critical directions.
- 4-bit offers the best size-accuracy balance; 2-bit is smaller with slight drops.
- We use general-purpose archivers as the final stage.
- Table: Residual+prune+4-bit reaches 61M with stable accuracy.

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Takeaways and Limitations/Outlook

Model	Size	Examination					Knowledge	Reasoning			Understanding			Avg
		C-Eval	CMMLU	MMLU	AGI-Eval	BoolQ	AX-b	PIQA	CSL	EPRSTMT	XSum	LCSTS		
PanGu- π -1B	Ori	14.64G	38.05	37.86	34.96	30.42	58.62	43.75	62.02	56.25	55.62	16.00	14.60	40.74
	Ours	0.59G	36.71	38.65	37.13	31.87	59.30	42.66	61.10	55.00	56.25	16.31	14.14	40.83
PanGu- π -7B	Ori	98.59G	59.91	59.97	61.84	54.04	64.59	56.88	77.31	63.12	90.00	19.59	16.61	56.71
	Ours	4.10G	61.32	60.14	62.37	55.11	68.44	52.90	77.91	63.75	90.00	19.24	16.77	57.09

- We enable extreme, near-lossless checkpoint compression via residuals, joint shrinking, and non-uniform quantization.
 - e.g., ~70 $\times$ on Pythia-410M; ~25 $\times$ on PanGu π -1B/7B
- Limitations: ratios vary with training regime and numeric format; very low bits trade small accuracy; assumes Adam-like optimizers and deterministic resumes.
- Outlook: extend to other optimizers/modalities and integrate more tightly with training infrastructure.
- Table: PanGu π —~25 $\times$ with negligible change.

Thank You

Questions & Discussion